



32548-48730 Cybersecurity
Week 1
Introduction to Cybersecurity



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Objectives



Welcome All!

In this Workshop I will present information on following topics:

- Subject organisation and class logistics
- Understanding the Goals of Cybersecurity
- Discuss on recent and ongoing Cybersecurity incidences
- Cybersecurity Threat, Attacks and Vulnerabilities
- Security approaches (mitigation strategy)

Subject Overview: Teaching Staffs



Teaching staffs

- Subject Co-Ordinator and Lecturer

A/Prof. Priyadarsi Nanda

Email: Priyadarsi.Nanda@uts.edu.au

- Workshop/Lecture Venue: CB 03.05.010
- Lab/Tutorial instructors (next page)
- Lab location CB11.06.102
- Remember, all lab/tutorial activities will be conducted face-to-face in the lab, NO online lab activities.

Get started			✓	+	⋮
⋮	📄	Welcome	✓		⋮
⋮	📄	Your teaching staff	✓		⋮
⋮	📄	Subject schedule	✓		⋮
⋮	📄	How to be successful in this subject	✓		⋮
⋮	📄	Assessment overview	✓		⋮
⋮	📄	Subject resources	✓		⋮
⋮	📄	Get started with UTS Library	✓		⋮
⋮	📄	Study help and support	✓		⋮
⋮	📄	Important guidance	✓		⋮

<https://canvas.uts.edu.au/courses/42130>

Weekly Activities



- 1.5-hour Workshop/Lecture + Live zoom Session, Wednesday, 18.00 – 19.30
 - Join URL: <https://utsmeet.zoom.us/j/81630207772>
 - Check your lab enrolment (Activity) and attend accordingly
- My Weekly consultation Time: Wednesday, 15.00-16.00, FEIT Learning Precinct, Building 11, Level 5
- If you want to meet personally, please email me at: Priyadarsi.Nanda@uts.edu.au
- For All Lab related matters, Contact your Lab instructor directly

48730:

Cmp1

01: Thursday 18.00-19.30: Ishan Nair
02: Thursday 19.30-21.00: Ishan Nair
04: Thursday 09.30-11.00: Furqan Siddiqui
05: Thursday 11.00-12.30: Furqan Siddiqui
07: Wednesday 10.30-12.00: Bashair Alrashed
08: Wednesday 12.00-13.30: Bashair Alrashed
09: Friday: 15.00-16.30: Swayam Tejas Padhy
10: Friday: 16.30-18.00: Swayam Tejas Padhy
11: Friday: 18.00-19.30: Md. Arif Hassan
12: Wednesday: 18.00-19.30: Md. Arif Hassan
13: Wednesday: 16.30-18.00: Shariful Siddique
14: Wednesday: 15.00-16.30: Shariful Siddique
15: Monday: 11.00-12.30: Abir Hasan
16: Tuesday: 09.30-11.00: Hrushikesh Dolas
17: Monday: 12.30-14.00: Abir Hasan

32548:

Cmp1

01: Wednesday 19.30-21.00: Md Arif Hassan
02: Thursday 12.30-14.00: Rishikesh Vijay
03: Thursday 14.00-15.30: Rishikesh Vijay

Workshop/Lecture and Lab materials



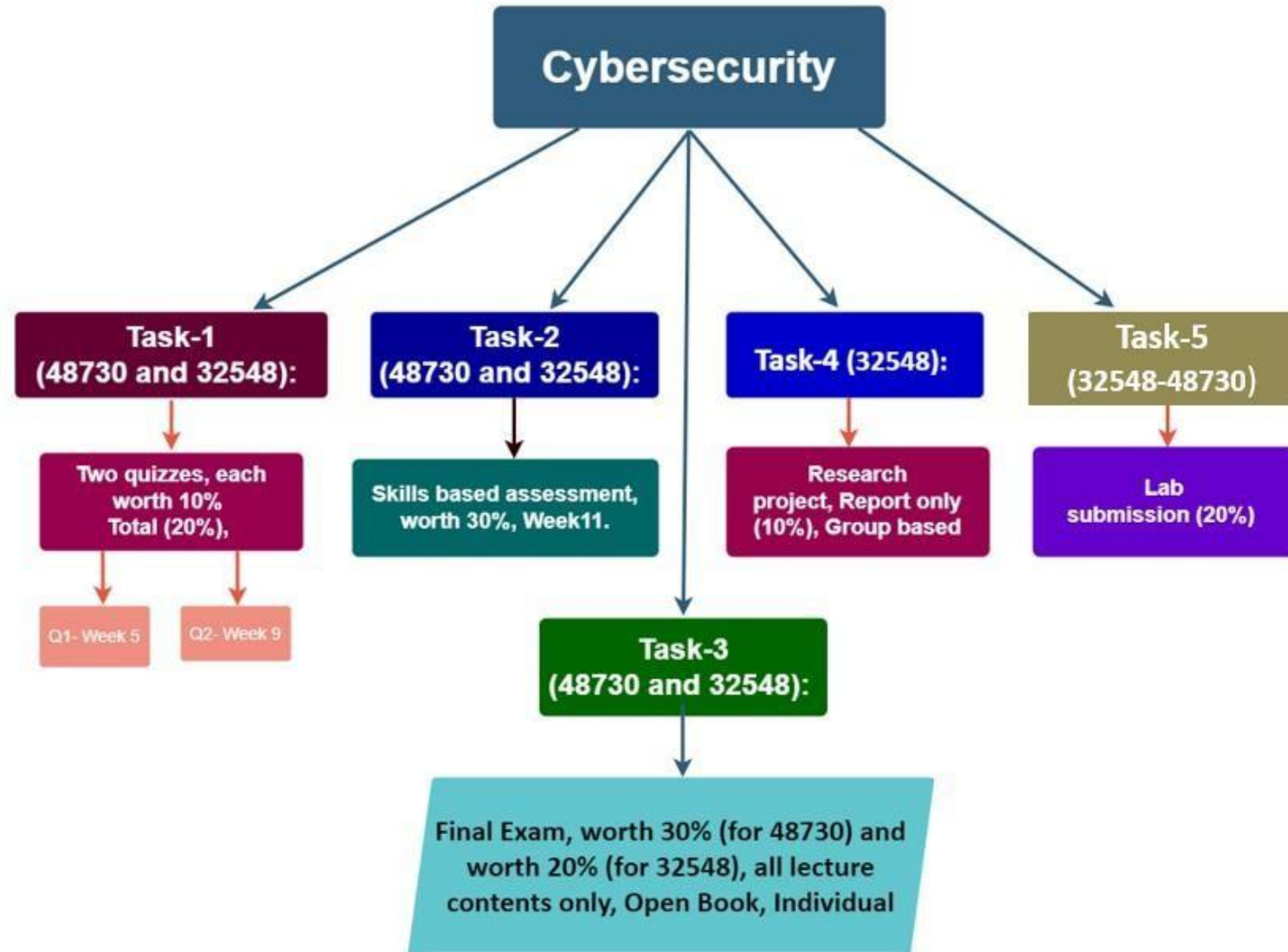
- All materials will be posted at **CANVAS**
 - Lecture and Lab materials will be available before the weekly workshop session
 - Lecture recordings will be posted after the workshop/lecture every week
 - Students enrolled in on-line workshop may access the workshop recordings at CANVAs modules
- Please **read** the materials before the workshop/lecture and lab session
- All additional reading materials (when required) will be posted at CANVAS
- Week-10, Week-11 are meant to be guest lectures
 - Guest Lecture contents will be examined

Assessments and Examination



- Task-1 (48730 and 32548): Two quizzes, each worth 10% (Total 20%), Only multiple-choice, multiple-answer, True/False type questions, **Closed Book, Individual**
 - Quiz-1: **Week-5**; will cover lecture materials from Week-1 to Week-4
 - Quiz-2: **Week-9**; will cover lecture materials from Week-5 to Week-8
- Task-2 (48730 and 32548): Skills based assessment, worth 30%, questions will be based on lab scenario and similar questions performed during the lab, **Closed Book, Individual**
 - Week-11
- Task-3 (48730 and 32548): Final Exam, worth 30% (for 48730) and worth 20% (for 32548), all lecture contents only, **Restricted Open Book, Individual**; **Use of AI tools are not permitted in this subject while answering the questions. Penalty applies.**
 - Short answer type questions
 - During UTS exam period
- Task-4 (32548): Research project, Report only (10%), **Group based**
- Task-5 (48730): Lab submission after the lab, through CANVAS and due date will be provided in CANVAS, 20%, **Individual**
- **Make sure you check CANVAS regularly on any updates!**
- **To pass the subject, a student must achieve an overall mark of 50% or more.**

Assessments and Examination



Tips to do Well in this subject



- Read your Lecture notes and attend workshop/lecture and listen to recordings
- Perform the lab exercises and submit your lab works timely
- Complete all assessment tasks
- Consult your subject coordinator
- Consult your Lab/Tutor instructor



Best of luck !!!

Pre-lab Activities



Week 1- Resources for Lab/Tutorial + Homework Lab 1 - Part 1

Hello everyone,

We will use Virtual machines for our labs from Week 2, please follow the "lab environment setup" to download the VMware workstation player and images.

On-campus lab activities will commence in Week 2.

Week 1 part 1 Tasks are for homework, the use of Virtual Machines is not required for this lab. You can find the instructions for Week 1 Lab (Information Gathering) in the Lab Handout below:

[Cybersecurity Week 1 - Lab 1, Part 1-2.pdf](#) ↓

Lab Environment Setup:

--> Sign into the Google Drive [Link for Lab Images](#) ↗ using your UTS account then downloads the VM files.

[Cybersecurity VM Lab Environment Setup-1.pdf](#) ↓

Vmware link - <https://www.vmware.com/info/workstation-player/evaluation> ↗

Or Download the installation file using your UTS account- https://drive.google.com/file/d/1fU_9GqZ7qlfNJPGFFESmYY8Rsle4KysG/view?usp=sharing ↗

NOTE: The lab computers are installed with VMWare Player 16 and VMWare Workstations.

It is highly recommended that you are ready with the VMware / Fusion Software installed on your personal devices and have the downloaded VM images ready for week2 lab. If you have any issues, please discuss them with your Tutor during the lab.

Additional information to setup Lab Environment using MAC computer (Not for M1 Chip based MAC):

[VM Lab Environment Setup - Mac-1.pdf](#) ↓

You may explore further on how to run VMWare on Apple M1 & M2 chip based computers. Microsoft recently releases this information. But, it is up to you to try.

<https://support.microsoft.com/en-us/windows/options-for-using-windows-11-with-mac-computers-with-apple-m1-and-m2-chips-cd15fd62-9b34-4b78-b0bc-121baa3c568c> ↗

Why Cybersecurity is important in today's context?



Cybercrime now costs more than natural disasters and most illegal drug trades combined

\$16.46T

Projected global cybercrime cost by 2030, rising from \$9.5T in 2024, nearly matching the world's 3rd largest economy

96%

Organizations lack cybersecurity confidence
Only 4% feel confident that connected device users are protected against cyberattacks

Critical sectors like healthcare, transport, and finance are prime targets with severe societal impact

\$5.12M

The average data breach cost in 2025 is the highest on record, up 5% from 2024. Healthcare breaches top \$10.3M.

Cyber Attacks Happen Every 39 Seconds!!!!!!

-Ransomware attack now happens every 15 sec or less

Sources: IBM 2025 Cost of a Data Breach | Verizon DBIR 2025 | ENISA Threat Landscape 2025 | Statista | ACSC Annual Cyber Report 2025 | University of Maryland; Clark School study; Security.org 2025

For more details- https://reports.weforum.org/docs/WEF_Global_Cybersecurity_Outlook_2026.pdf

Why Cybersecurity is important in today's context? [2]



68%
of breaches caused
by human error; the
weakest link

1,876
attacks/week per
organization (75%)

Only 15%
have adopted Zero
Trust — most rely on
outdated perimeter
defenses

75%
faced
ransomware (avg.
\$5.1M per hit)

2,200+
Cyberattacks
daily — someone
falls victim every
39 sec

4,151%
surge in AI-driven
phishing (since
ChatGPT boom)

277 days
Average time to
detect & contain
a breach

107%
rise in IoT malware
attacks

Cybersecurity threats **are not theoretical**; they are **happening right now, all the time**.

Sources: IBM 2025 Cost of a Data Breach | Verizon DBIR 2025 | ENISA Threat Landscape 2025 | ACSC Annual Cyber Threat Report 2025 | Statista 2025 | University of Maryland; Clark School Study | Security.org 2025

What is Cybersecurity?



- **Cybersecurity is the practice of defending information systems**, including *hardware, software, networks, and data*. From unauthorized access, disruption, or destruction by malicious actors.
- **Ensures the core principles (CIA):**
 -  Confidentiality — information is only accessible to authorized users
 -  Integrity — information is accurate and trustworthy
 -  Availability — systems and data are accessible when needed
- Security goals can be achieved by combining:
 -  People: skilled professionals and informed users
 -  Processes: policies, governance, incident response plans
 -  Technologies: firewalls, encryption, intrusion detection, monitoring etc.
- Evolving business Focus:
 - ✓ Zero Trust – “Never trust, always verify”; assumes breaches happen, continuously validates users & devices.
 - ✓ Cyber Resilience – ensures quick recovery & continuity after attacks.



Modern security = CIA + Zero Trust + Resilience + AI

Cybersecurity vs. Information Security vs. Data Security



- **Sharing information in many forms**
- **Information Security (InfoSec)**
 - Covers all forms of information, not just digital
 - Includes physical (paper), digital files, software, networks
- **Data Security**
 - Focused on protecting valuable data and information assets
 - Ensures data is not stolen, corrupted, or lost
- **Cybersecurity**
 - Most critical aspect in today's connected world
 - Protects almost everything digital:
 - Computers, servers, mobile devices, networks
 - Critical infrastructure systems (energy, transport, healthcare)
 - Cyber-physical systems & IoTs



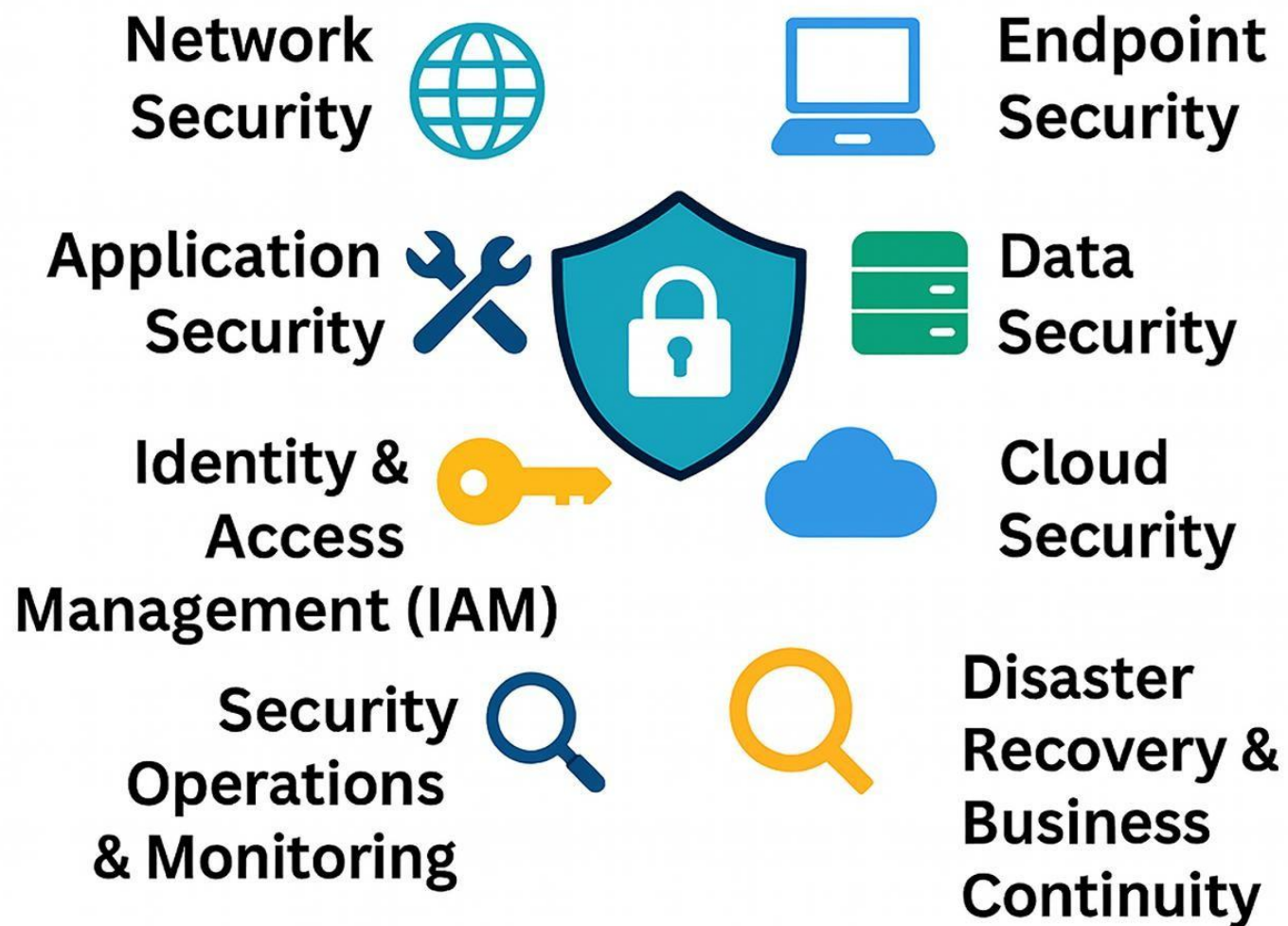
All interconnected, but, each with a different focus & scope

Introduction: Understanding Cybersecurity



- Cybersecurity is young, but fast evolving due to ongoing risks faced by organizations, countries and individuals
- Cyber incidences are growing almost exponentially!
- A list of resources for you to understand Cyber incidences
 - <https://www.techtarget.com/searchsecurity/feature/Ransomware-trends-statistics-and-facts>
 - <https://gemcell.com.au/news/cyber-threats-2026/>
 - <https://www.csis.org/programs/strategic-technologies-program/significant-cyber-incidents>
 - <https://purplesec.us/security-insights/data-breaches/>
 - <https://www.csis.org/programs/strategic-technologies-program/significant-cyber-incidents>
 - <https://www.pwc.com.au/cyber-security-digital-trust/global-digital-trust-insights.html>
 - <https://www.cyber.gov.au/>
 - <https://www.dst.defence.gov.au/sites/default/files/publications/documents/Future-Cyber-Security-Landscape.pdf>
 - <https://www.bleepingcomputer.com/news/security/the-biggest-cybersecurity-and-cyberattack-stories-of-2025/>
 - How success we are in **Detecting and Preventing** Cyber-attacks?

Key Domains of Cybersecurity



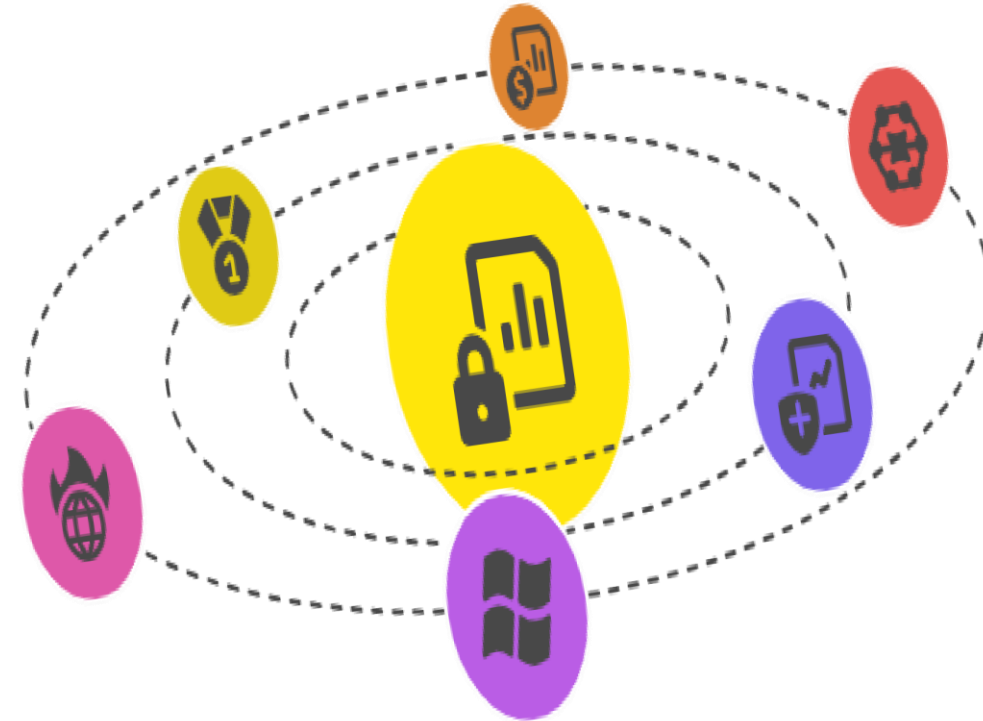
Cybersecurity in 2025-2026: Key Industry Threat Insights



Premium Threat Intelligence and Cybersecurity Vendor Reports:

These leading vendor reports provide deep dives into real-world incidents, financial impacts, and evolving attacker tactics in 2025-2026.

- [Verizon Data Breach Investigations Report \(DBIR\)](#)
→ Gold standard for breach statistics and attack patterns
- [IBM Security Cost of a Data Breach Report](#)
→ Detailed financial impact analysis and recovery metrics
- [Mandiant M-Trends Report](#)
→ APT (Advanced Persistent Threat) analysis and attribution
- [CrowdStrike Global Threat Report](#)
→ Real-time threat landscape analysis
- [Microsoft Digital Defense Report](#)
→ Global threat landscape from Microsoft's perspective
- [Sophos Zero Trust Network Access \(ZTNA\)](#)
→ Enforcing secure access for SMB and enterprise



Together with public sources, these reports form a robust picture of the global cyber threat landscape.

Cybersecurity in 2025-2026: Essential Tools & Response Platforms

- [**SANS Internet Storm Center**](#)
→ Daily security diary and incident tracking
- [**VirusTotal**](#)
→ Malware analysis and incident investigation
- [**Shodan**](#)
→ Search engine for internet-connected devices, helps discover exposed systems & vulnerabilities
- [**YARA Rules Repository**](#)
→ Malware detection rules for incident investigation
- [**Volatility Framework**](#)
→ Memory forensics for incident investigation
- [**The Sleuth Kit \(TSK\)**](#)
→ Digital forensics and incident analysis tools
- [**ISC2 \(International Information System Security Certification Consortium\)**](#)
→ CISSP: Certified Information Systems Security Professional
- [**Cybrary**](#)
→ Free cybersecurity training, including incident response
- [**SANS Cyber Aces**](#)
→ Free foundational cybersecurity tutorials



Cyber Attack Incident Diagram



Why use live threat maps? They help demonstrate:

- ✓ The constant, global scale of attacks
- ✓ The speed & volume of threats
- ✓ Why security is an *ongoing process, not a one-off project*

Cyberattacks don't sleep! Always monitoring = early detection

Explore these live maps:

- [Kaspersky CyberMap](#)
- [Checkpoint ThreatMap](#)
- [Fortinet Threat Map](#)
- [Netscout DDoS Map](#)
- [Bitdefender Threat Map](#)
- [Radware Live Threat Map](#)



⚠ Note: They're not 100% comprehensive, but they give you a clear sense of just how **constant and global** cyber attacks are.

Common Cyber Attacks?



⚙️ **Malware & Supply Chain Attacks**

Malware hidden in downloads or vendor software, which then gives them access deeper inside organizations.

👤 **Phishing & social engineering**

It could be a fake email, a text, or even a voice deepfake.

🧠 **Cloud & SaaS Misconfigurations**

Misconfigured cloud services , like leaving a storage bucket public or overly broad IAM permissions or giving apps too many permissions.

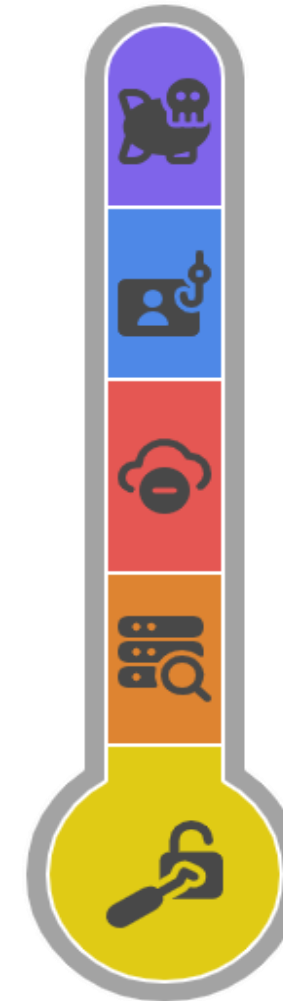
💻 **Exploiting Unpatched Systems**

Attackers scan the internet looking for outdated systems they can easily break into.

🔑 **Stolen or Weak Credentials**

Take advantage of stolen or weak passwords, often reused passwords from old breaches, or by guessing simple passwords.

Complex



Malware

Hides in downloads, vendor software

Phishing

Uses fake emails, texts, deepfakes

Cloud Misconfigurations

Exploits public buckets, excessive permissions

Unpatched Systems

Scans for outdated, vulnerable systems

Stolen Credentials

Exploits weak passwords, old breaches

Simple

Motives Behind Cyber Attacks & Their Huge Implications



Damaging Trust, Security, and Privacy to:

- 💰 Gain financial advantage, through theft, ransom, fraud.
- ❑ 🏢 Undermine **competitors' brand reputation**.
- ❑ ☠️ Spread fear via **cyberterrorism** and sabotage.
- ❑ 🕵️ Steal **sensitive government** & business secrets.
- ❑ ⚔️ Engage in cyber warfare targeting **critical systems**.

Massive Implications After an Attack

- ❑ 📉 Operational **shutdowns & huge** recovery costs.
- ❑ 🔒 **Loss of customer trust** & long-term loyalty.
- ❑ 🚨 Legal & regulatory fallout (fines, investigations)
- ❑ 📰 Damaging headlines that **erode market confidence**.

Some Common origin of Cybersecurity threats



Most cyber incidents start with simple mistakes

Phishing & Suspicious Emails

- Trick users into clicking malicious links or attachments.

Weak Passwords & Credential Reuse

- Easy targets for brute-force & credential stuffing.

Outdated Software & Systems

- Unpatched vulnerabilities are exploited by attackers.

Unverified Third-Party Apps

- May install hidden malware or backdoors.

Poor Security Hygiene & Human Error

- Lack of awareness, ignoring security warnings.

Lack of Multi-Factor Authentication (MFA)

- Single point of failure, easier account takeover.



Which of these risks do you think your organization is most vulnerable to?

Data Breaches Report: 2025-2026 Australia



Victorian Department of Education-January 2026

- [Government and education sector-Unknown threat actor](#) | Student names, school emails, year levels, and encrypted passwords were accessed by an unauthorised party

Spectrum Medical Imaging – January 2025

- [Sydney medical practice cyber incident claimed by INC Ransom](#)

Evidn – January 2025

- [Everest ransomware gang lists Aussie company Evidn as a victim](#) | Hackers claim to have stolen 50 gigabytes from an applied behavioural science firm that works closely with the Queensland government.

Volkswagen – January 2025

- [Almost 800k Volkswagen EV owners data exposed](#)

SquareX – January 2025

- [SquareX reveals critical breach of Cyberhaven extension](#) | SquareX has revealed a critical browser security incident targeting Chrome Extension developers, leading to a major compromise of Cyberhaven's browser extension.

MediSecure – January 2025

- [Company at centre of data breach revealed](#)

More details



<https://www.webberinsurance.com.au/data-breaches-list#twentyfive>

<https://borderlesscs.com.au/2026-data-breach-lists>

Security Strategy: Strong Foundations



Detection

- Spot suspicious activities, policy violations & unknown threats.
- Use SIEM, threat hunting & EDR tools to catch intrusions early.



Prevention

- Harden systems: patches, MFA, encryption, zero trust networks.
- Train staff to block phishing & social engineering.



Response & Recovery

- **Contain breaches fast, limit spread & data loss.**
- **Restore operations, learn from incidents, improve defences.**



Cybersecurity actions range from proactive to reactive measures.

Constant vigilance is the **best defense**, stay ahead of threats with real-time intelligence. Cybersecurity is the backbone of the **digital economy** & trust in society.

The CIA Triad — Core Security Goals



Confidentiality

- ❑ **Goal:** Prevent unauthorized disclosure of data.
- ❑ **What it protects:** Intellectual property (IP), personal IDs, financial & health records, trade secrets.
- ❑ **Typical breaches:**
 - Stolen laptops & devices
 - Phishing or social engineering attacks
 - Insider leaks & unauthorized access

Integrity

- ❑ **Goal:** Ensure data & systems are accurate, trustworthy, unaltered.
- ❑ **Covers:**
 - Data integrity: data only changes in authorized ways.
 - System integrity: systems function as intended without tampering.
- ❑ **Threat examples:**
 - Malware altering data
 - Unauthorized database edits
 - Fraudulent transactions injected by hackers

Availability

- ❑ **Goal:** Keep systems & data accessible to authorized users when needed.
- ❑ **Risks include:**
 - DDoS attacks crashing services
 - Hardware failures or sabotage
 - Failed backups causing data loss



Additional Critical Security Goals [1]



Trusted Users

Authorized individuals



Secure Access

Controlled entry to systems

🔑 Authenticity

Goal: Ensure users are who they claim, data comes from a trusted source.

Real-world scenarios:

- 🐟 Phishing emails pretending to be your bank
- 🌐 Fake websites mimicking legit services
- 👤 Social media impersonation attacks

Methods:

- 📄 Digital signatures & certificates
- 🗝️ Two-factor authentication (2FA)
- ☐ Biometric checks (fingerprint, face ID)
- 🔑 Public Key Infrastructure (PKI)

🏢 Access Control

Goal: Only authorized people/processes access

Core principles:

- 🔍 **Need-to-know:** Minimum necessary access
- ⚖️ **Separation of duties:** No one person has total control
- ♻️ **Regular reviews:** Remove unused permissions

Methods:

- 👤 Role-based access control (RBAC)
- 🗝️ Multi-factor authentication (MFA)
- 🚦 Least privilege principle
- ⌚ Time-based access restrictions

Additional Critical Security Goals [2]



Accountability & Non-Repudiation

Goal:

Ensure actions are traceable to specific individuals or entities, preventing denial after the fact.

Why it matters:

-  Legal compliance & forensic investigations
-  Stops “I didn’t do it” claims
-  Meets regulatory requirements

How it’s implemented:

-  Digital timestamps & cryptographic signatures
-  Detailed audit trails
-  Blockchain

Example:

Audit logs prove who changed sensitive data.

Detection & Monitoring

Goal:

Quickly spot security threats & incidents as they occur.

Key activities:

-  Real-time system & network monitoring
-  Intrusion Detection Systems (IDS)
-  SIEM (Security Info & Event Management)
-  Behavioral analytics & anomaly detection

Monitoring targets:

-  Network traffic patterns
-  Login attempts & user behaviors
-  File system changes
-  System performance metrics

Example:

Alerts on repeated failed logins, unusual data access, or suspicious network traffic.

Security vulnerability, penetration testing and security threats



What is a Vulnerability?

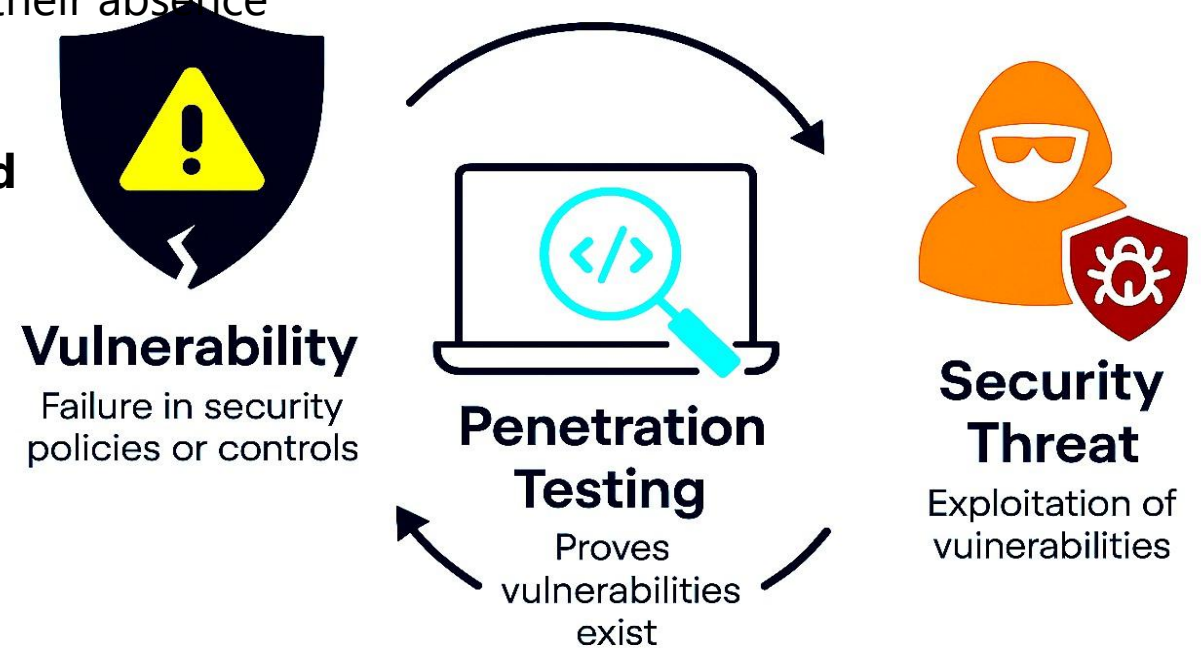
- A gap in security policies, processes, or controls
- Discovering vulnerabilities in the systems attackers perform unauthorized actions
- Examples: unpatched software, misconfigurations, weak passwords

□ Penetration Testing (Pentest):

- A method to prove **the presence of vulnerabilities**, not their absence
- Simulates real attacks to discover weak points

When Threats Happen:

- A threat becomes real when **vulnerabilities are exploited**
- Leads to data theft, system downtime, or financial loss



Why it matters:

Knowing vulnerabilities is the first step to fixing them!



information gathering or footprinting or reconnaissance

Passive Reconnaissance

- Passive reconnaissance involves acquiring information without directly interacting with the target
- For example, searching public records or news releases

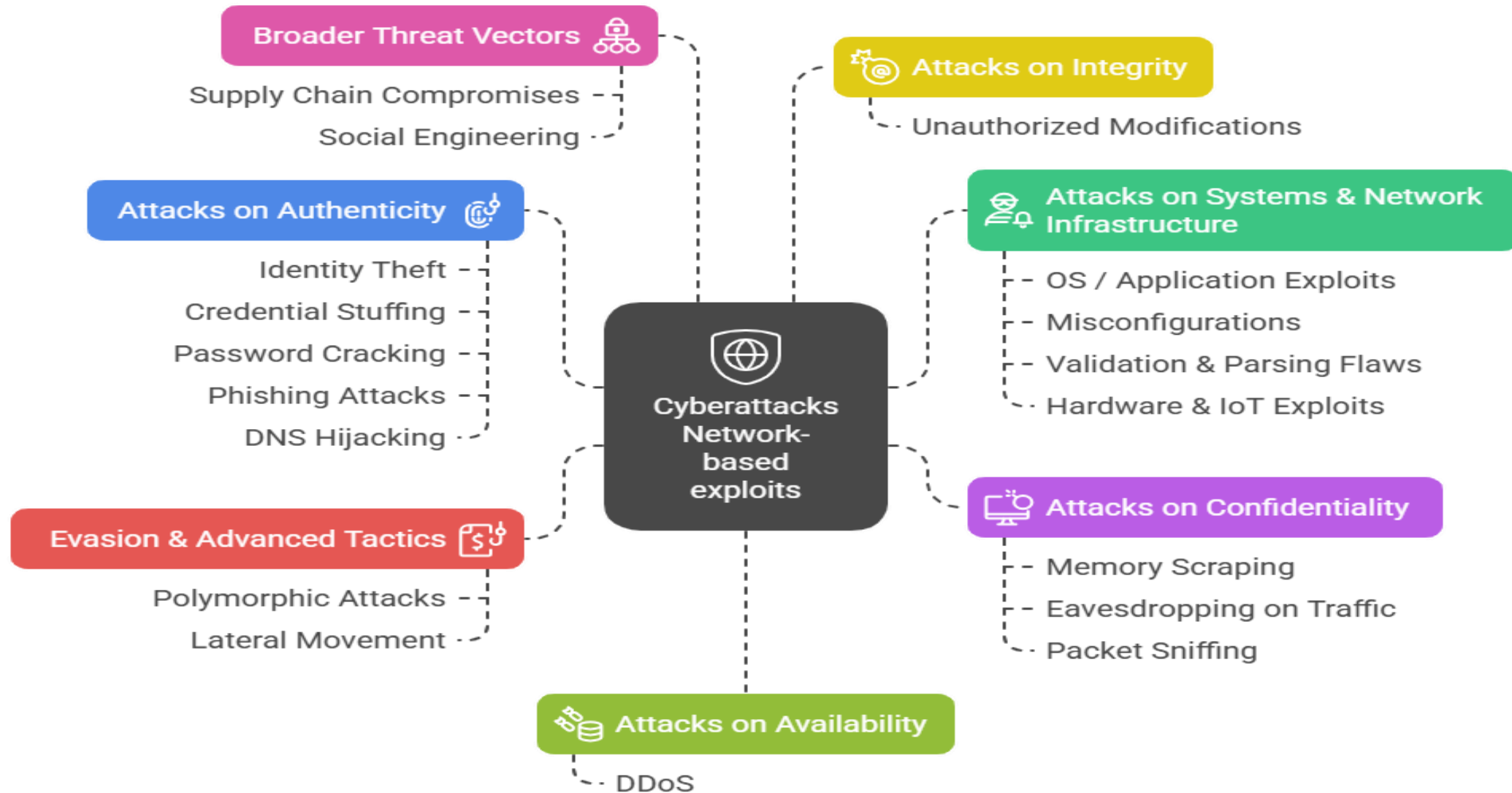
Active Reconnaissance

- Active reconnaissance involves interacting with the target directly by any means
- For example, telephone calls to the help desk or technical department

A photograph of the UTS (University of Technology Sydney) building at night. The building is illuminated, and several bright green laser beams are projected from the ground towards the facade. The UTS logo is visible on the upper right part of the building.



Cyber attacks: Network-based exploits



Regular patching, secure configurations, network segmentation, & staff awareness are key to reducing the impact & spread of these attacks.

List of security vulnerabilities



🔌 Injection & Parsing Flaws

- SQL, NoSQL, Command Injection, buffer overflows

🔑 Broken Authentication & Sessions

- Weak login, brute force, poor MFA

🚧 Access Control Failures

- Privilege escalation, Insecure direct object references (IDOR)

🔑 Cryptographic Failures

- Weak encryption, bad key management

⚙️ Security Misconfigurations

- Default creds, exposed cloud services

📁 Sensitive Data Exposure

- Leaking PII, secrets in logs

❑ Unsafe Deserialization & Object Handling

- Leads to remote code execution

📡 Vulnerable & Outdated Components

- Unpatched libraries, firmware

🌐 API & Server-side request forgery (SSRF) Weaknesses

- Over-permissive or unauthenticated APIs

🔭 Insufficient Logging & Monitoring

- Missed intrusions, no anomaly alerts

😬 Cross-Site Scripting (XSS) & Content Injection

- Reflected, stored, DOM XSS

🕷️ Race Conditions & Memory Safety

- Use-after-free, concurrency bugs

Human error and security attacks



Input & Validation Flaws:

- ☐ Unsanitized Inputs: Not filtering special characters (SQLi, XSS)
- ☐ Blind Trust in Inputs: Not validating external data/components
- ☐ **Poor Handling & State Issues:**
- ☐ Poor State Management: Losing track of session or data integrity
- ☐ Uncaught Errors: Failing to handle exceptions securely
- ☐ Race Conditions: Failure to secure concurrency in critical operations

Weak Cryptography & Auth:

- ☐ Weak Cryptography: Outdated algorithms, bad key practices
- ☐ Weak Authentication: Poor password policies, no MFA
- ☐ Weak Authorization: Role or access checks missing or broken

Network & Data Transport Gaps:

- ☐ Unprotected Network Traffic: Missing TLS, no VPN

Data Protection Failures:

- ☐ Insecure Data Handling: Storing data unencrypted or misclassified

Poor Security Design Choices:

- ☐ Security by Obscurity: Relying on hidden mechanisms instead of strong controls
- ☐ Outdated Dependencies: Using software with known CVEs



Severe Security Vulnerabilities - 2025 Statistics



<i>Vulnerability</i>	<i>Impact</i>	<i>Source</i>
Zero-day exploits	28.3% exploited within 1-day of disclosure	https://vulncheck.com/blog/exploitation-trends-q1-2025
Supply chain attacks	45% of organizations will experience by 2025	https://www.ivaniti.com/blog/software-supply-chain-attack-risk
Cloud vulnerabilities	154% increase in 2024	https://www.embroker.com/blog/top-cybersecurity-threats/
Mobile app vulnerabilities	75% of apps contain at least one vulnerability	https://touchlane.com/common-mobile-application-vulnerabilities-2025/
CVE growth	22,254 CVEs (30% increase)	https://www.indusface.com/blog/key-cybersecurity-statistics/
Linux vulnerabilities	967% increase in 2024	https://thehackernews.com/2025/05/weekly-recap-zero-day-exploits-insider.html
Ransomware attacks	8% increase across North America in 2025	https://cybersecuritynews.com/2025-cybersecurity-trends/
AI-powered attacks	67% increase compared to 2024	https://cybersecuritynews.com/2025-cybersecurity-trends/
Software supply chain	25% increase in attacks	https://thecyberexpress.com/software-supply-chain-attacks-have-surged/
Unpatched vulnerabilities	60% of breaches caused by unpatched systems	https://www.indusface.com/blog/key-cybersecurity-statistics/

Some Graphical presentation of Attackers?



Professional criminals
Make money from attacks



Amateur hackers
Fun and attention



Government hackers
Steal secrets, disrupt enemies



Industrial espionage specialists
Market manipulation



Hired attackers
Specific target



Company employees
Revenge or personal gain



Elite hacker teams
Long-term spying



Excitement hunters
Adrenaline and bragging rights



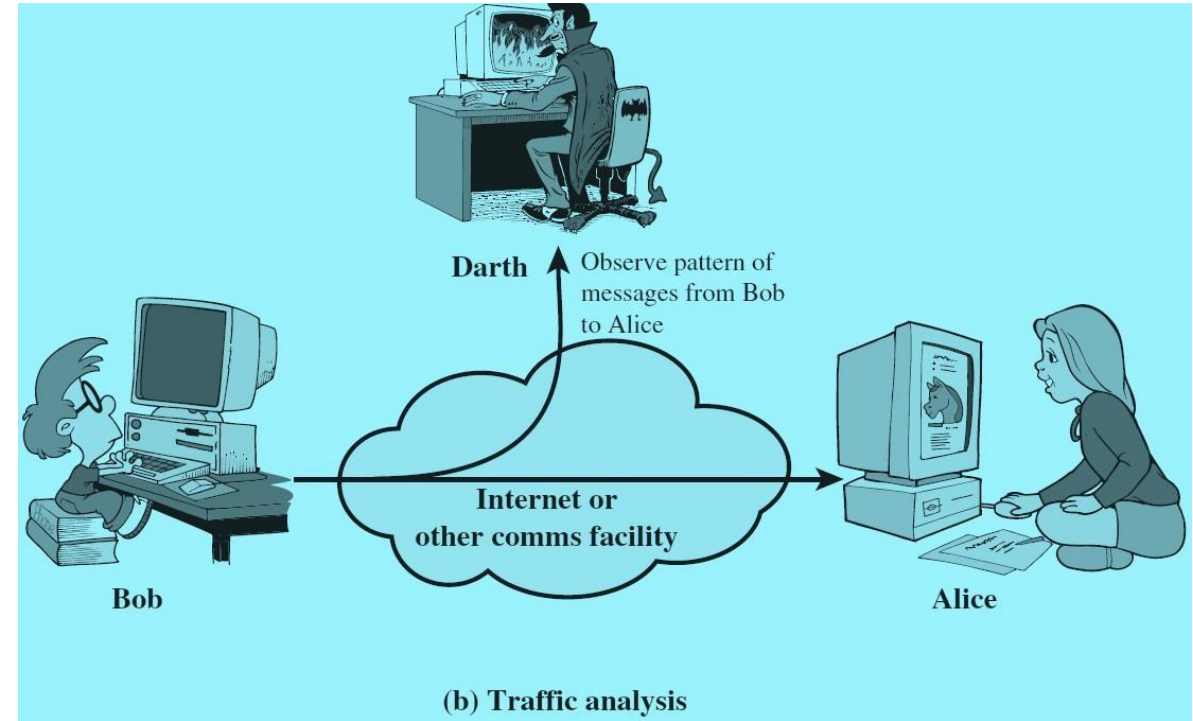
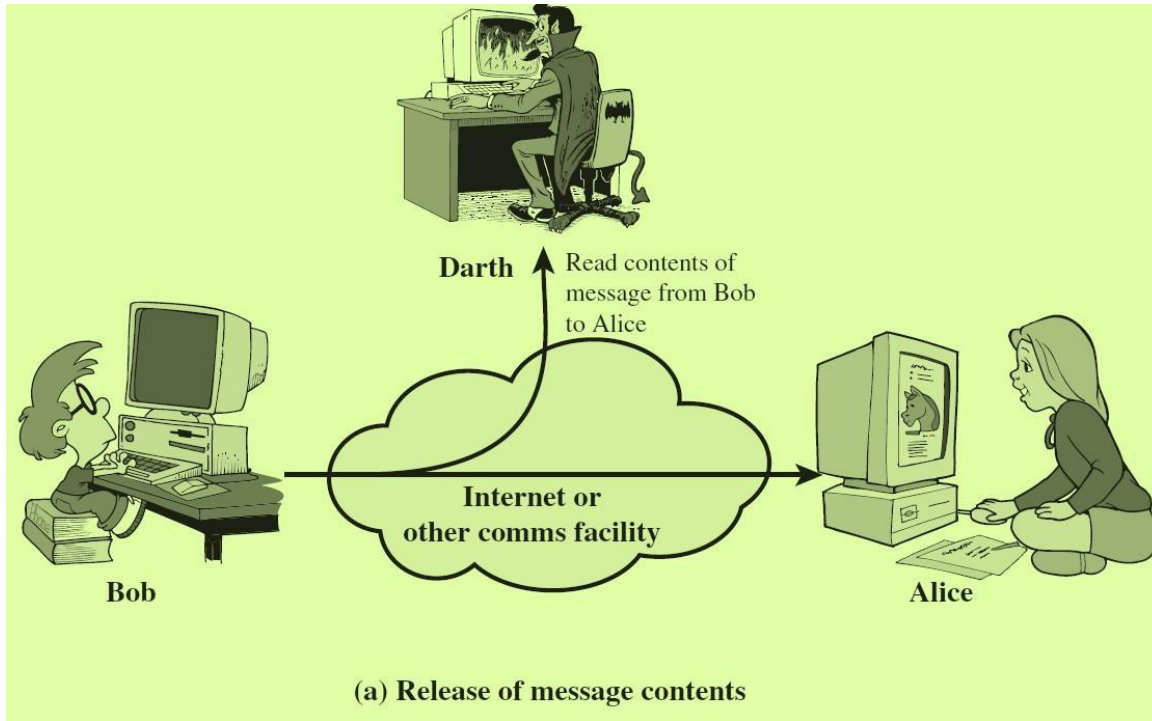
Political protesters
Promote their cause



Ransomware gangs
Hold data for ransom

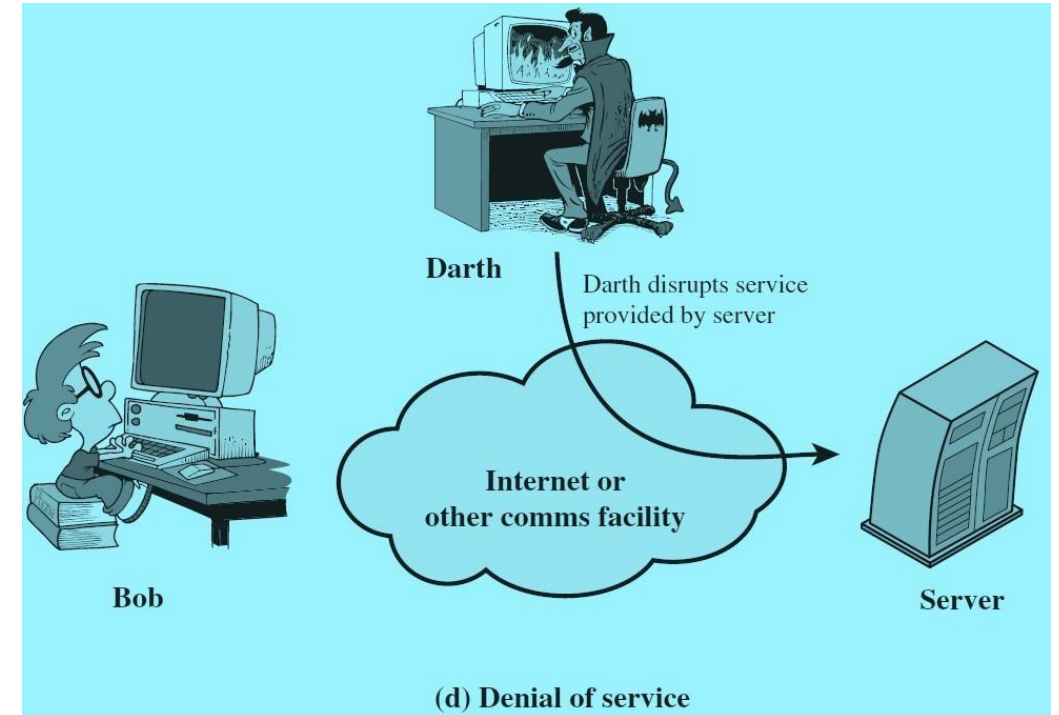
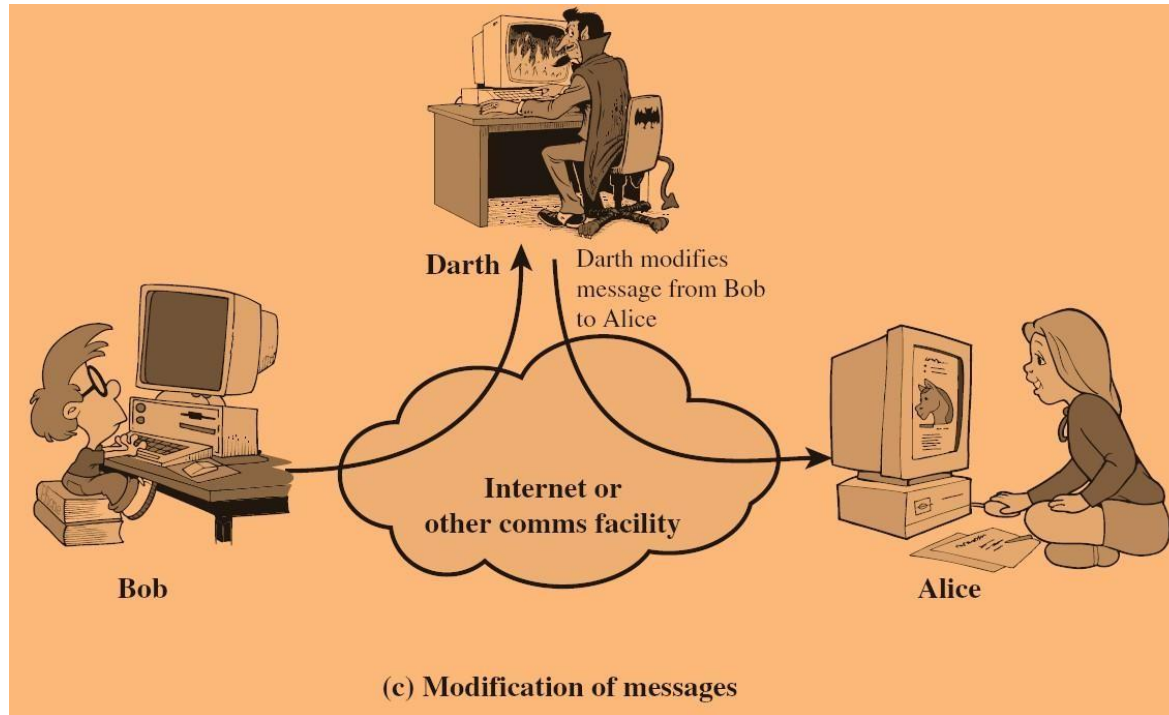


A Simple Attack Scenario



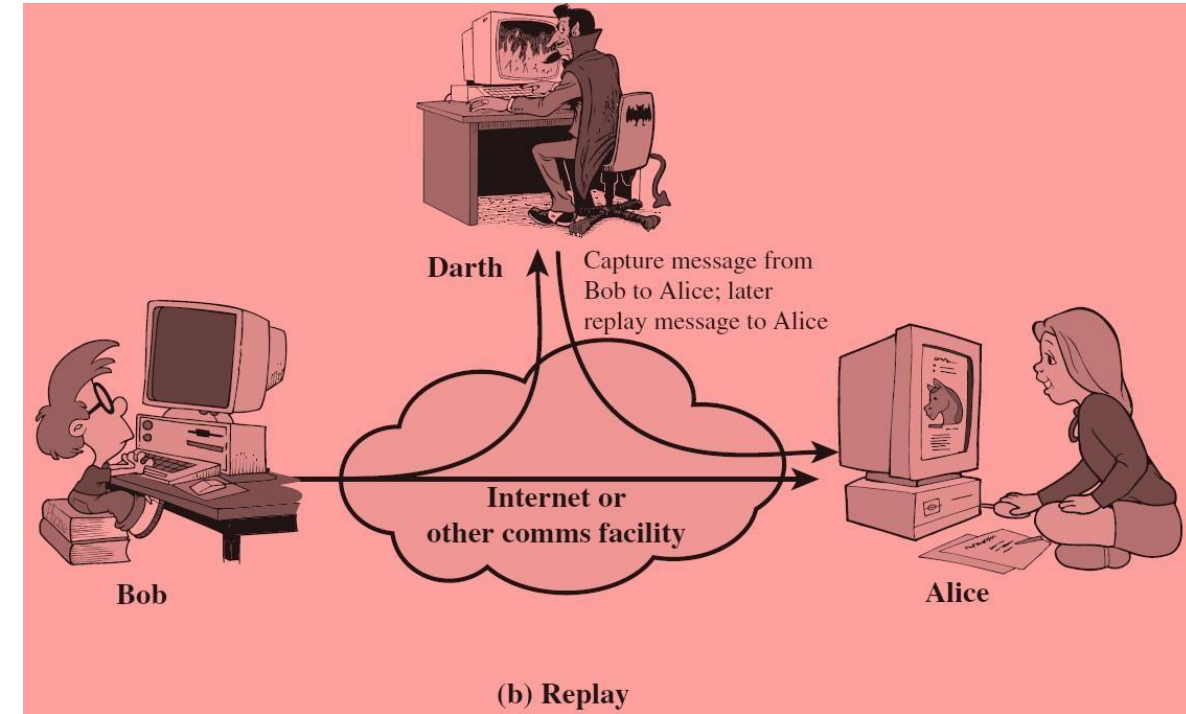
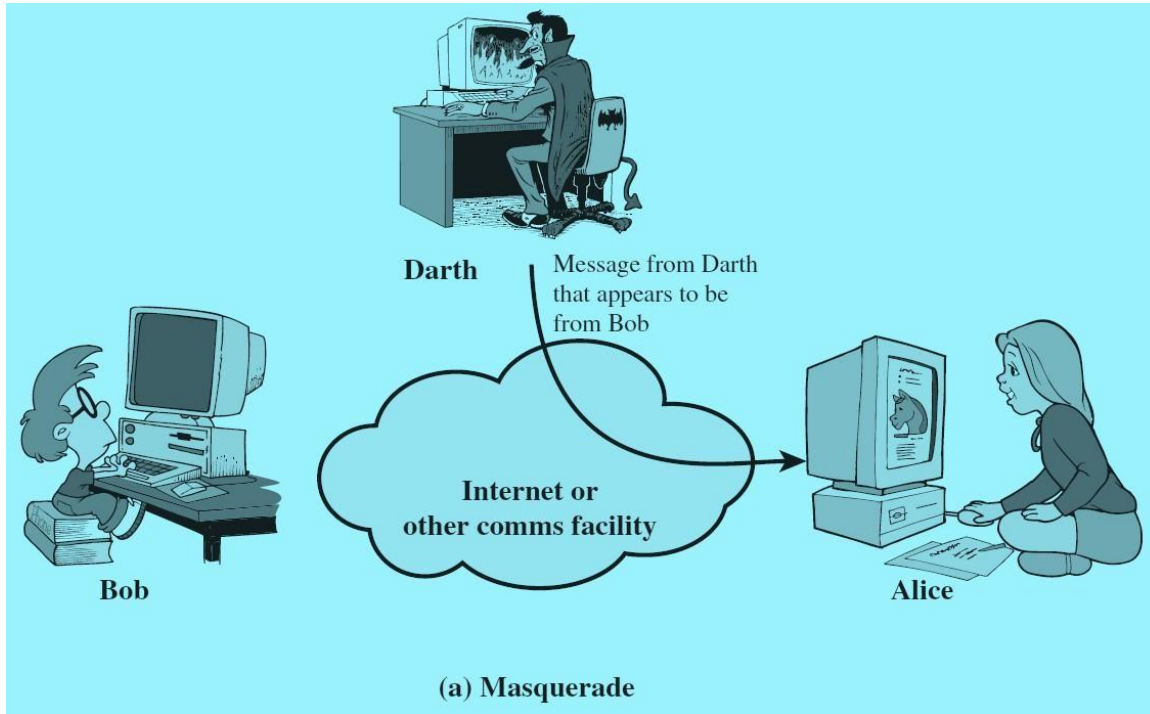
Source: "Network Security Essentials: Applications and Standards", 4th Edition by William Stallings

Attack Scenario Cont.



Source: "Network Security Essentials: Applications and Standards", 4th Edition by William Stallings

Attack Scenario Cont.



Source: "Network Security Essentials: Applications and Standards", 4th Edition by William Stallings

What are the challenges in Cybersecurity?

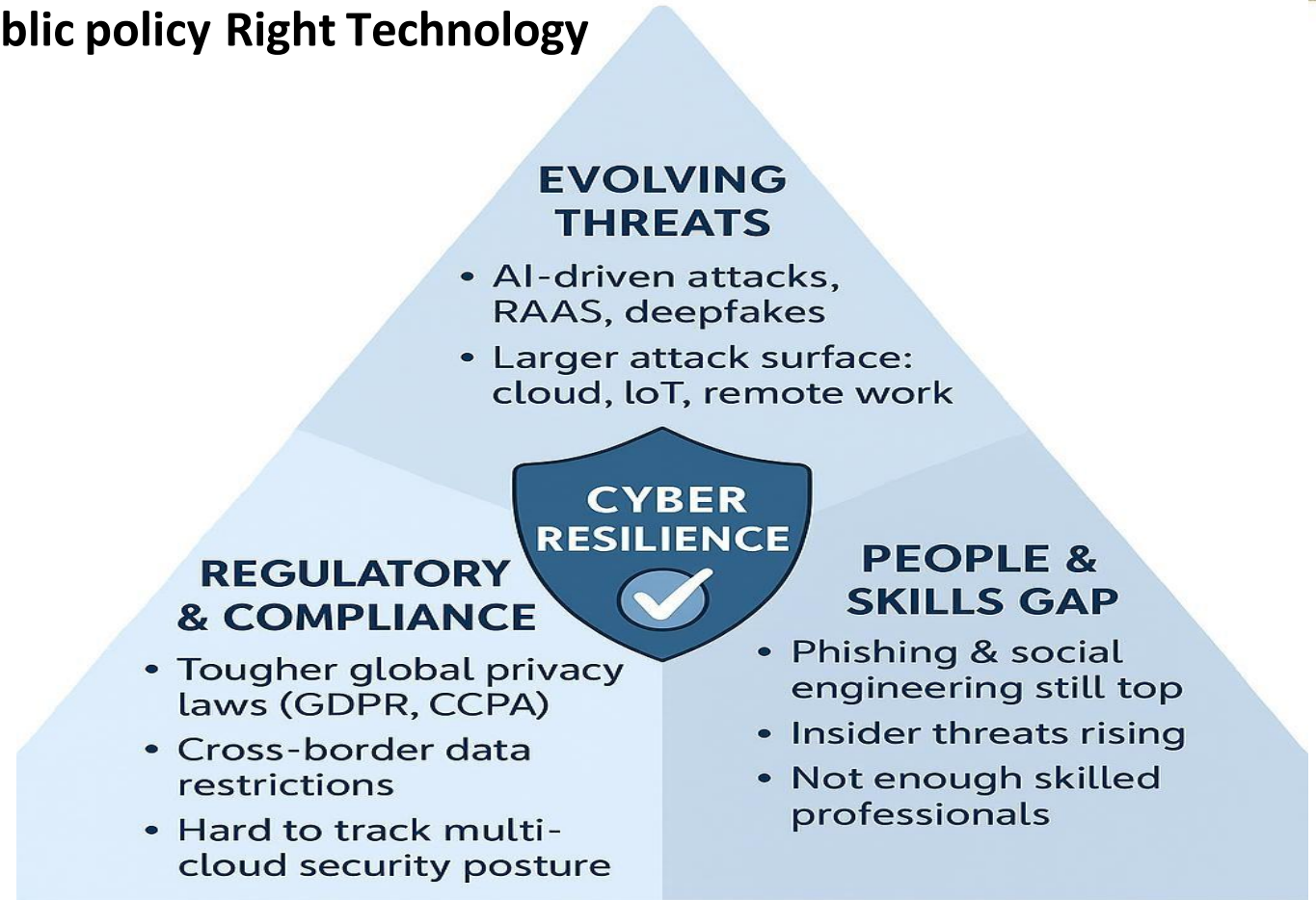


Combination of right Technology and right Public policy Right Technology

- ☐ What, When and Where to apply?
- ☐ Simple Vs. Complex
- ☐ Policy is crucial
- ☐ User awareness is equally important

Cybersecurity hurdles:

- There are no metrics measuring in-security
- Internet has no boundary
- No limit on how many device can be connected to Internet?



In 2026, no single tool or policy is enough. Cyber resilience demands aligned tech, people & policy or attackers will find the weakest link.

Attacks on industrial control systems (ICS)



Surge in ICS Ransomware Attacks

► 87% increase in attacks on industrial sectors in 2024–2025, crippling manufacturing & energy operations.

Norway Dam Hack

► Cyberattack manipulated dam control systems, exposing risks to water infrastructure.

Iberian Peninsula Grid Outage

► Major blackout across Spain & Portugal linked to suspected cyber intrusion into power management.

Targeted Iranian Campaigns

► U.S. DHS warns of coordinated attacks on water plants & power facilities by Iranian threat actors.

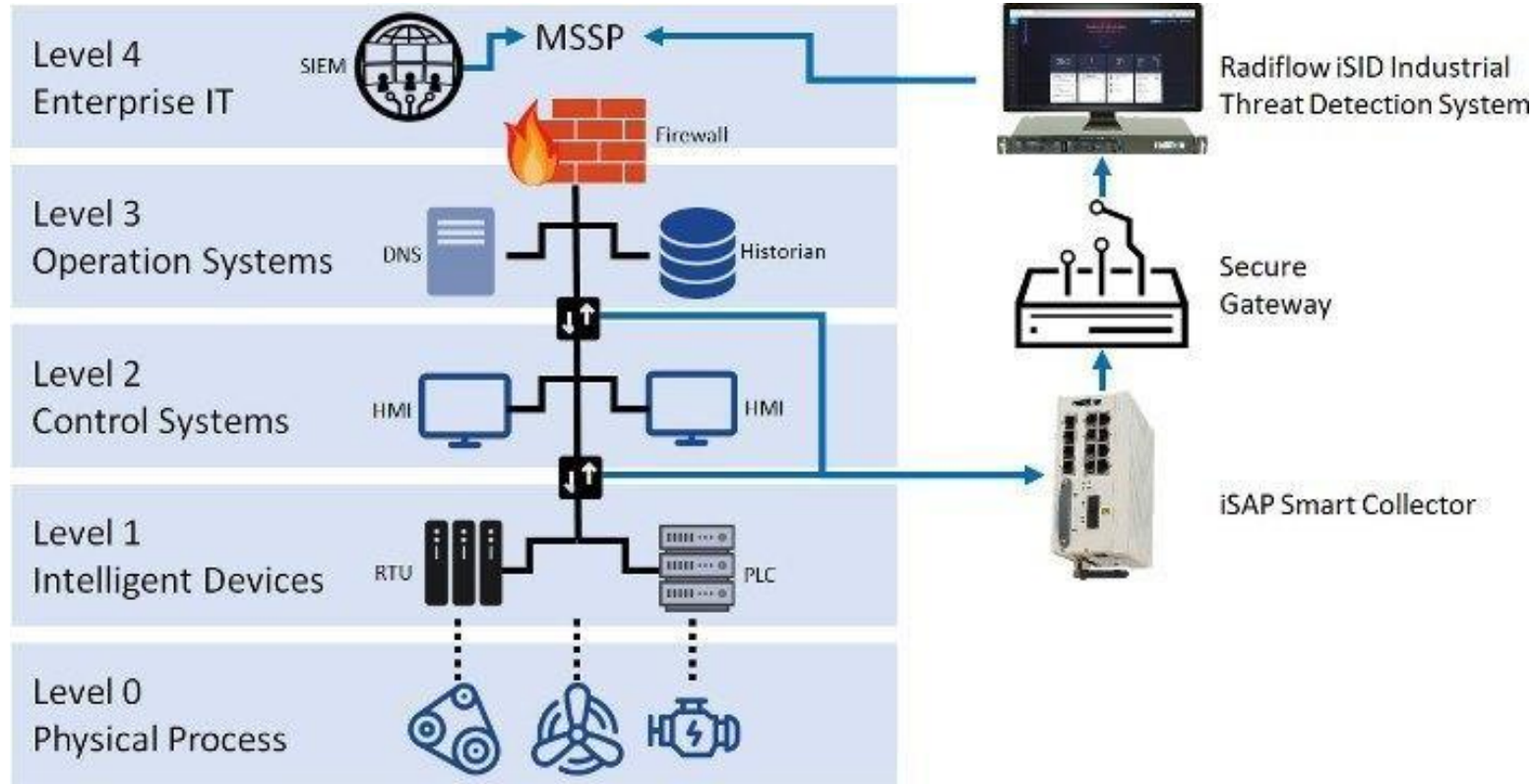
Industry 4.0 Expands Attack Surface

► Growth of IIoT devices, AI, and real-time automation deepens dependency & heightens exposure.



✓ As IT merges with OT, industrial environments face not only data theft but also **operational disruption & safety risks**, elevating cyber threats to a national security level.

IIoT Security Architectures & Threat Mitigation Approaches



Other sources exploring ICS & IIoT security architectures:

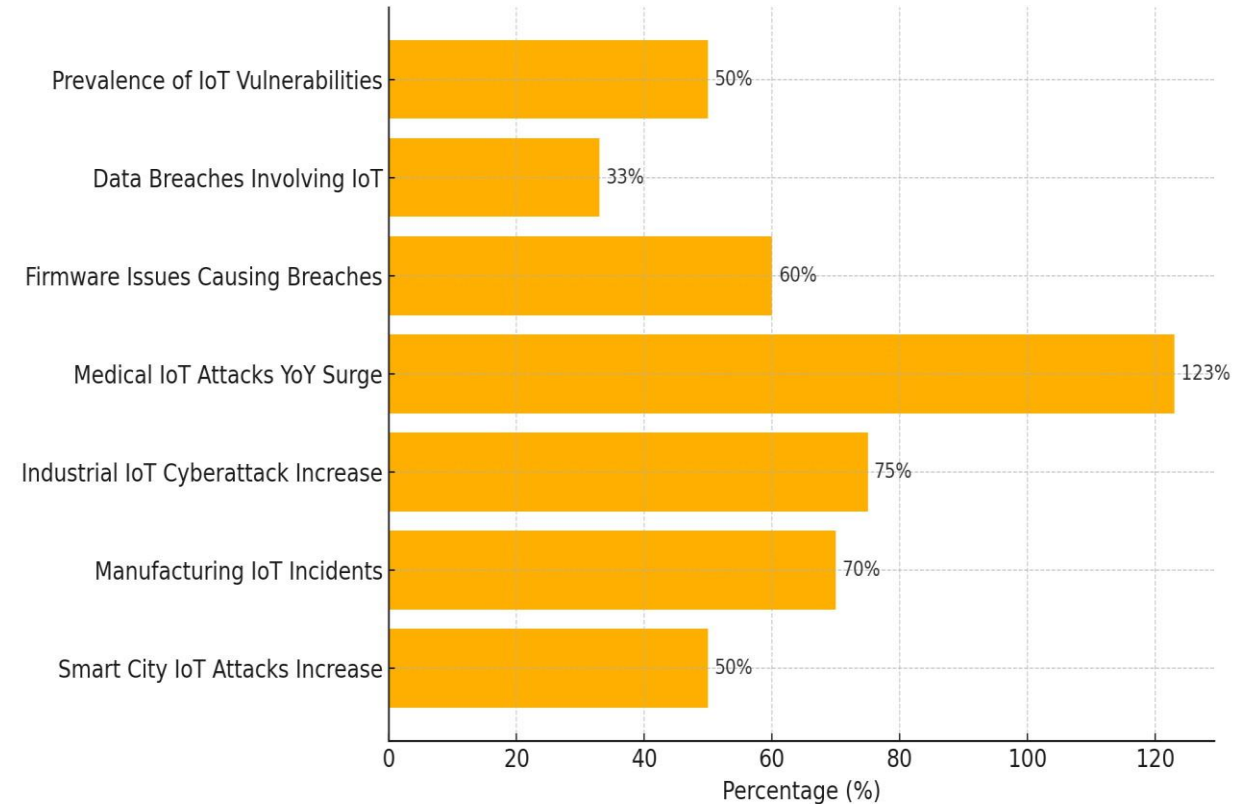
- 🔗 [Claroty](#) — OT/ICS asset visibility, risk management & segmentation
- 🔗 [Nozomi Networks](#) — ICS & IoT anomaly detection, asset inventory
- 🔗 [Dragos](#) — OT threat intelligence & incident response frameworks
- 🔗 [Siemens SINEC](#) — Industrial network management, patch & secure remote access
- 🔗 [Cisco OT Security](#) — Network segmentation, VLANs, Zero Trust for OT

<https://radiflow.com/ot-mssp-partner-program/>

Smart Home Usage & IoT Security Risks



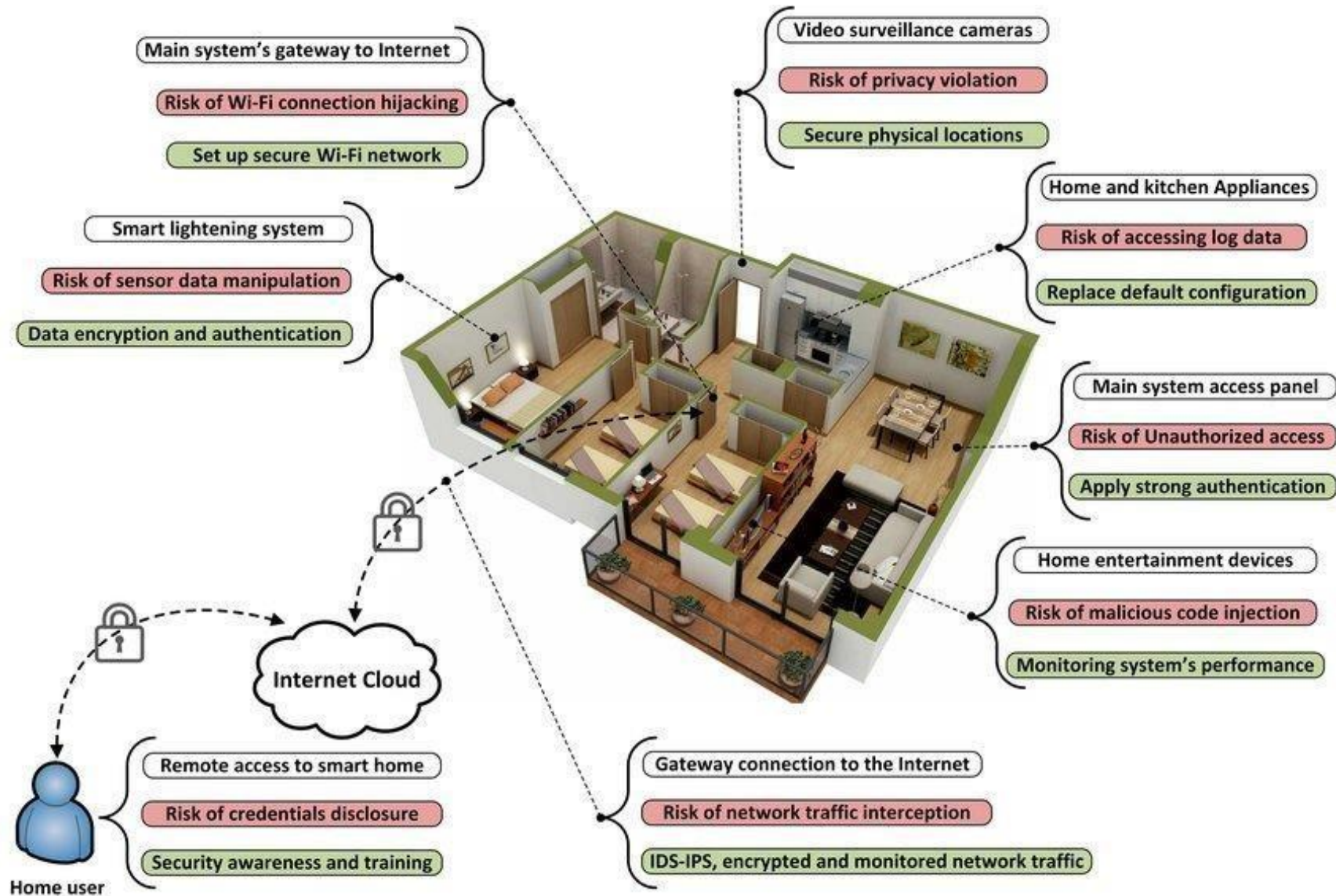
Survey Methodology: An online survey of 1,000 nationally representative U.S. adults ages 18+.



<https://jumpcloud.com/blog/iot-security-risks-stats-and-trends-to-know-in-2025>

<https://www.statista.com/forecasts/887613/number-of-smart-homes-in-the-smart-home-market-in-the-world#:~:text=The%20global%20indicator%20'Smart%20Homes,a%20new%20peak%20in%202028>

Smart Home Device Threats



The Security Landscape



Recent trends in cyber incidents

- **Attack severity increasing:** About **42% of reported vulnerabilities now receive CVSS scores above 7.0**, indicating more critical risks than five years ago.
- **Broader attack surface:** AI & IoT are driving higher exposure across industries, with machine identities outnumbering humans **80:1** (CyberArk 2025).

Key Threat Landscape Reports

- ENISA <https://www.enisa.europa.eu/>
- Microsoft www.microsoft.com/sir
- IT Governance - <https://www.itgovernance.co.uk/blog/global-data-breaches-and-cyber-attacks-in-2024>
- Kaspersky Lab Security report: <https://securelist.com/>
- 2024 Identity Security Threat Landscape Report: <https://www.cyberark.com/threat-landscape/>
- [Annual Cyber Threat Report: https://www.cyber.gov.au/sites/default/files/2023-03/ACSC-Annual-Cyber-Threat-Report-2022_0.pdf](https://www.cyber.gov.au/sites/default/files/2023-03/ACSC-Annual-Cyber-Threat-Report-2022_0.pdf)
- Also, many more organizations have been publishing their own reports, Look through those reports.

Pick any two reports, summarize the top trends, and discuss with peers.



Common Vulnerability Scoring System (CVSS) Calculator

Rating	CVSS Score
None	0.0
Low	0.1-3.9
Medium	4.0-6.9
High	7.0-8.9
Critical	9.0-10.0

- NIST outlined
 - Method of calculating vulnerability score
 - <http://nvd.nist.gov/cvss.cfm?calculator&version=2>
- Mitre corporation mentioned a number of products for vulnerability assessment (CVE: Common Vulnerabilities and Exposures)
 - https://cve.mitre.org/compatible/vulnerability_management.html
- Security Requirements, Federal Information Processing Standards (FIPS 200)
 - <https://nvlpubs.nist.gov/nistpubs/FIPS/NIST.FIPS.200.pdf>

The Top 4 Mitigation Strategies # 1



1 Application Whitelisting

- Only allow **trusted, validated software** to run.
- Helps block malware from executing.

2 Regular Patching & Vulnerability Management

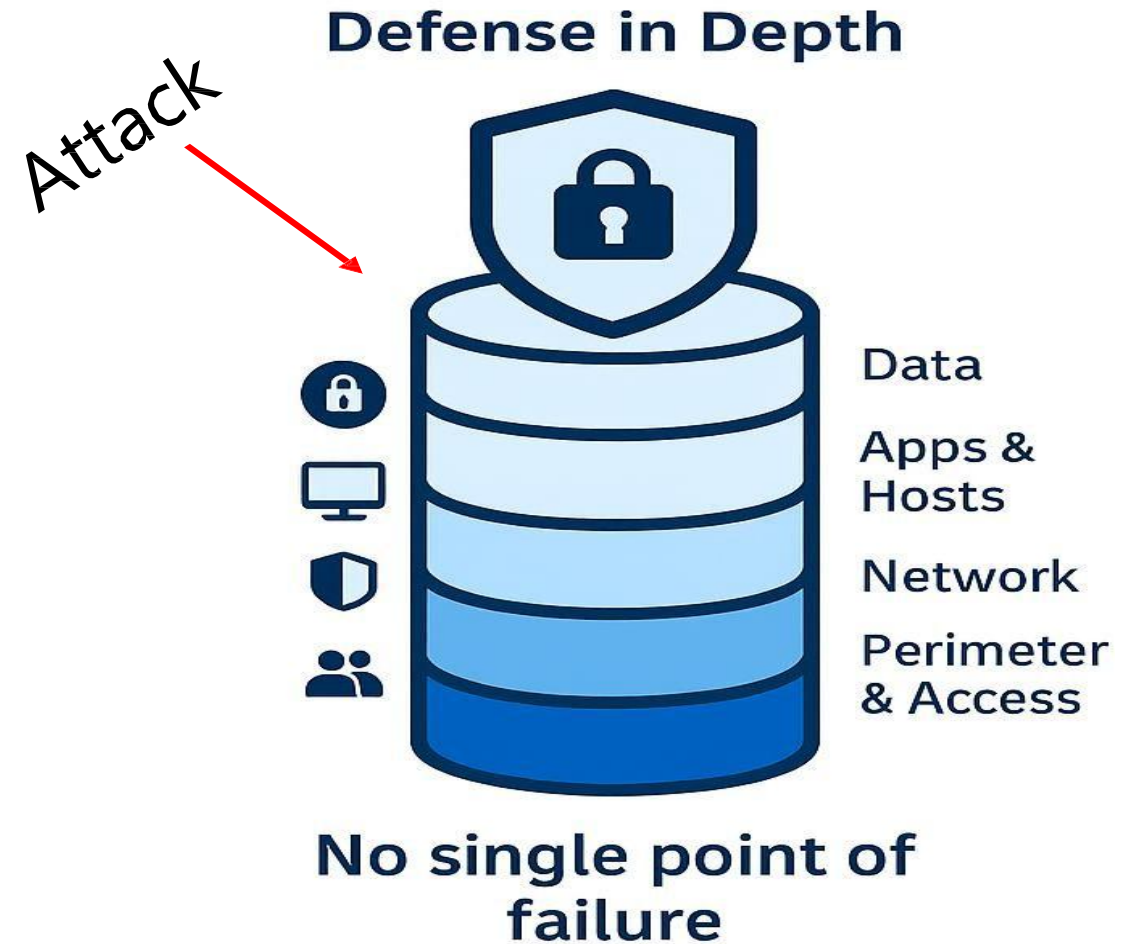
- Automate updates across OS & apps.
- Monitor patch failures, run vulnerability scans.

3 Restricting Administrative Privileges

- Use **least privilege** principle (no internet from root accounts).
- Apply MFA for all admin access.

4 Building a Defense-in-Depth System

- Layered protections across:
 -  **Data** (encryption, backups)
 -  **Applications & Hosts** (EDR, allowlists)
 -  **Networks** (firewalls, segmentation)
 -  **Perimeter & Access** (VPNs, Zero Trust gateways)
 -  **User Layer** (policies, security awareness)



How to stay cyber safe?



- Cyber hygiene
- Assess and manage potential risks
- Determine security metrics and use them efficiently
- Strategy to recover from cyber incident
- Be aware and constantly educate using various resources

User awareness and Education contributes significantly maintaining a secured environment

Readings, Textbook and references



- Stallings: Chapter 1
- Chwan-Hwa Wu and David Irwin: Chapter-17
- Internet Resources as mentioned throughout the slides
- William Stallings, "*Network Security Essentials Applications and Standards*" (5th Edition/6th Edition) ISBN-10: 0133370437, ISBN-13: 978-0133370430.
- Chwan-Hwa (john) Wu & J.David Irwin, "*Introduction to Computer Networks and Cyber Security*", CRC Press, ISBN: 978-1-4665-7213-3.
- Wenliang Du, "*Computer and Internet Security, A Hands-on approach*", Second Edition, ISBN: 978-1-7330039-3-3
- Michael E. Whitman & Herbert J. Mattord, "*Principles of Information Security*", 7th/8th Edition, ISBN: 978-1-337-10276-6, Chapters 1, 2, 5.
- William Stallings & Lawrie Brown, "*Computer Security: Principles and Practice*", 4th Edition, ISBN: 978-0-13-479410-5, Chapters 1, 4, 12.
- Charles J. Brooks et al., "*Cybersecurity Essentials*", ISBN: 978-1-119-36375-2, Chapters 3, 6.
- Bernard Menezes, "*Network Security and Cryptography*", ISBN: 978-0-470-82466-6, Chapters 1, 7.
- Michael T. Simpson, Nicholas Antill, "*Hands-On Ethical Hacking and Network Defense*", ISBN: 978-1-337-10204-9, Chapters 2, 11.
- Joseph Migga Kizza, "*Guide to Computer Network Security*", 6th Edition, ISBN: 978-3-030-37937-5, Chapters 4, 8.
- There are other recommended texts mentioned in your Subject Outline
- **Note: I will also use materials from other resources and will be advised to you during the semester**